

DEFENSE & DISASTER ROBOTICS

● SWARM STATE: 5/5 ACTIVE

— AUTONOMOUS MULTI-AGENT AERIAL SWARM ARCHITECTURE

PROJECT SUTRA

Swarm Unified Tactical Reconnaissance Architecture

A decentralized, fault-tolerant multi-UAV swarm engineered from first principles for **GPS-denied** environments, **heavy RF jamming (-5 dB SNR)**, and sub-meter survivor geolocation under dense forest canopy.

SUBSYS A

PX4 50Hz Offboard + SUTRA-FSD 3D Occupancy & ORCA 3D

SUBSYS B

Deep JSCC

SUBSYS C

Tri-Modal RGB/Thermal Edge AI + 3.59cm WGS84 Raycaster

SUBSYS D

Pegasus 3D
CoT

232/232 TESTS

ZERO-MOCK VERIFICATION

100% automated pytest suite passing with live telemetry stdout

96.9% BW SAVED

DEEP JSCC COMPRESSION

512 KB → 16 KB frames surviving extraction